Fiscal Year:	FY 2021	**Task Last Updated:**	FY 03/18/2024
PI Name:	Chatterjee, Anushree Ph.D.		
Project Title:	Nucleic Acid Therapy Platform for Real-Time Countermeasures During Spaceflight Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80303-1904	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	01/01/2019	**End Date:**	09/30/2021
No. of Post Docs:	3	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2021 per E. Urquieta/TRISH (Ed., 5/17/21) NOTE: End date changed to 4/30/2022 per TRISH (Ed., 5/26/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nagpal, Prashant Ph.D. (University of Colorado at Boulder) Zea, Luis Ph.D. (University of Colorado at Boulder)		
Grant/Contract No.:	NNX16AO69A-T0406		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In long space exploration missions, astronauts are subjected to increased amounts of chronic radiation exposure, which can lead to Acute Radiation Syndrome (ARS) causing DNA damage, tissue damage, and aging. The goal of the project is to develop a high-throughput, customizable, gene-specific therapeutic strategy that will allow targeting of any gene for countermeasures, the design of an on-site development, and deployment of therapy for astronauts in need. Recent research efforts using nucleic acid therapy of mammalian cells include microRNAs, small interfering RNAs, long non-coding RNAs, deactivated CRISPR-Cas9 technology; however, though promising, these therapeutic strategies involve tedious cloning and optimization that are time-consuming, require specific skills that may not be available aboard International Space Station (ISS). The proposed research offers a drastic new approach with the development of a Facile Accelerated Specific Therapeutic (FAST) pipeline to design, build, and test treatment within a time scale of hours to days against any mammalian gene of interest by outlining the following tasks. (Aim 1) Facile and Accelerated high throughput design of peptide nucleic acids (PNAs) based antisense therapeutics that can block translation of any desired gene by binding to its mRNA for targeted inhibition. (Aim 2) Automated and high throughput parallel synthesis of PNA therapeutic library for generation of on-site therapeutic molecules, and development of nanoparticle as well as bacterial Type III secretion system based probiotic system to efficiently deliver therapeutic PNA molecules to host cells to overcome challenges of transport, toxicity, and bioavailability. (Aim 3) Testing of designed and built PNAs in a high throughput manner in a relevant macrophage, neutrophil, and hematopoietic stem cells based models, as well as a macrophage-infection model in presence of gamma-radiation. The proposed method will allow identification of important gene targets, and quickly generate translatable therapies that can be tested under host conditions, and deployed on-site within a time scale of hours to days. It is expected that the FAST pipeline can be deployed in the future in spaceflight settings.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	EARTH AND SPACE: Development of platform technology to design countermeasure therapeutics rapidly to target almost any gene involved in radiation response. This can be valuable for cancer treatments. Demonstration of gene-specificity, reduced radiation response, increased transport, the lowered toxicity of PNAs. Novel nucleic acid/peptide purification and delivery strategy that can significantly impact broader fields of medicine with the capability to build disease and patient specific therapeutics. Development of novel therapeutics at an accelerated pace (< 1 week) that can be beneficial for broad range of countermeasures (such as new disease outbreaks). SPACE: Possibilities of deploying FAST platform on International Space Station (ISS) to provide radiation countermeasures (as well as other countermeasures--such as new antibiotics to treat drug-resistant infections). Automated high throughput PNA synthesis will reduce the cost of production as well as provide a method of PNA synthesis on-site (space station), therefore removing the need for long-term storage. The nanotechnology approach will provide a delivery vehicle for enhanced PNA transport, lowered toxicity, and increased bioavailability.
Task Progress:	In long space exploration missions, astronauts are subjected to increased amounts of radiation exposure, which can lead to Acute Radiation Syndrome (ARS) causing DNA damage, tissue damage, and aging. Immune dysfunction associated with spaceflight conditions can also potentially increase the susceptibility of crew members to pathogenic bacteria in extended space. We have developed peptide nucleic acid-based therapeutics to develop countermeasures and preventative agents for relieving ARS with potential applications in future space missions. Our process is high-throughput, and generates customizable, gene-specific therapeutics with high precision that targets genes relevant for radiation countermeasures in human cell lines, by outlining the following steps. 1) Development of a bioinformatics toolbox for design of nucleic acid-based modulators of gene expression of targeted genes. 2) Automated and high throughput synthesis of nucleic acid-based expression modulator library. 3) Testing of designed and built nucleic acid-based modulators in a high throughput manner in relevant primary human cell lines (in presence and absence of space-relevant radiation). The irradiated cells have been screened for a panel of functional outcomes with known clinical relevance, including cytokine, protein and gene expression patterns. Importantly, our GM-CSF targeting analogs are able to confer a protective immune response upon low-dose gamma radiation exposure. Further collaborations (courtesy Sachi Bioworks), show that modified versions of the molecule are safe and effective in animal models. Using in vitro and in vivo models, we have shown that with improved delivery approaches (courtesy Sachi Bioworks), the technology can be also used for creating antivirals that are effective against SARS-CoV2. Our studies show that nucleic-acid therapeutics can effectively and safely regulate immune response, and thus provide a direct approach for radiation countermeasure that could be potentially deployed in future flight mission to serve medical needs for astronauts.
Bibliography Type:	Description: (Last Updated: 10/31/2025)
Articles in Peer-reviewed Journals	McCollum CR, Bertram JR, Nagpal P, Chatterjee A. "Photoactivated indium phosphide quantum dots treat multidrug-resistant bacterial abscesses in vivo." ACS Appl. Mater. Interfaces. 2021 Jun 22;13(26):30404-19. https://doi.org/10.1021/acsami.1c08306 ; PMID: 34156817 , Jun-2021
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